

SARAWAK MEDCHAIN

Sarawak's Sovereign Health Record System

Blockchain-secured · Patient-controlled · Audit-ready

BUSINESS PLAN

Version 1.0

Prepared · June 2026

PREPARED FOR

Cradle Fund Sdn Bhd

Sarawak Digital Economy Corp.

TEGAS · Prospective funders

CONFIDENTIAL · DRAFT v1.0

AT A GLANCE

THE ASK · AT A GLANCE

RM 50,000 – RM 150,000

Cradle CIP Spark / MyStartup · 12-month runway to first paying customers

TARGET BY MONTH 12

5 clinics + 2 hospitals

RM 42,500 MRR · RM 510k ARR

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This document is prepared for prospective funders, government partners, and pilot facilities. The live product may be verified any time at sarawak-medchain.pages.dev.

EXECUTIVE SUMMARY

Snapshot

Sarawak MedChain is a blockchain-secured medical certificate platform that eliminates MC fraud in Malaysia through cryptographic verification, patient-controlled access, and audit-ready reporting for state agencies.

The product is **live** at sarawak-medchain.pages.dev, with a **working OTP demo** at [/otp](https://otp) that lets doctors access patient records via a 6-digit code — no wallet, no app install required.

On **21 June 2026**, the Minister of Health, Datuk Seri Dr Dzulkefly Ahmad, publicly announced KKM is *studying e-MC implementation to curb misuse of sick leave certificates* (Astro Awani). Sarawak MedChain is a built, working implementation of exactly this — ready for a Sarawak pilot today.

STAGE

Pilot-ready · Pre-revenue

Enterprise-registered · Sdn Bhd upon first contract

THE ASK

RM 50,000 - 150,000

Cradle CIP Spark / MyStartup · 12-month runway to first paying customers

SECTION 1

The Problem

Malaysia loses an estimated **RM 1.5 - 3 billion** annually to medical certificate fraud. Paper-based MC systems offer zero protection:

- **Easily forged** - paper templates freely available online
- **Impossible to verify at the point of use** - employers cannot check authenticity
- **No audit trail** - no way to track issuance, access, or misuse
- **No central registry** - no way to confirm whether a specific MC was actually issued at the named facility, by the named doctor, on the claimed date

Estimate methodology: Malaysia healthcare spend (~RM 60B, MOH 2023) multiplied by the global healthcare fraud range of 3-7% (NHCAA - National Health Care Anti-Fraud Association). MC-specific data is not publicly tracked.

Market timing

On **21 June 2026**, Minister of Health Datuk Seri Dr Dzulkefly Ahmad publicly stated:

“KKM teliti pelaksanaan e-MC, kekang penyalahgunaan sijil cuti sakit”

- reported by Astro Awani (@501awani)

This is a Tier-1 public signal that Malaysia is actively moving toward e-MC. The addressable market is not hypothetical; it is being defined by the Health Ministry in real time.

SECTION 2

The Solution

Sarawak MedChain is a blockchain-secured medical certificate platform with three novel design decisions:

1. Client-side encryption (AES-256-GCM)

The medical record is encrypted in the browser *before* it leaves the doctor's device.

2. Public smart contract for cryptographic hashes

Only tamper-proof fingerprints go on-chain; personal data stays encrypted off-chain on IPFS (Malaysia-hosted).

3. Dual access modes

Wallet sign-in for hospital staff, and a 6-digit OTP for patients, elderly users, and guest doctors who need immediate access without app install.

How it works - 5 steps

01	Doctor issues Verified doctor signs in with secure wallet, fills the MC, submits.
02	Encrypted & hashed AES-256-GCM client-side; hash on public smart contract.
03	Patient controls Grant or revoke access to specific doctors/employers anytime; every access logged on-chain.
04	Employer verifies Anyone scans the QR code; verified in under 5 seconds.
05	Audit-ready Every action logged on-chain; state agencies see anonymised, aggregated data.

Live product URLs

- **Landing:** sarawak-medchain.pages.dev
- **Working OTP demo:** sarawak-medchain.pages.dev/#/otp
- **Government preview dashboard:** sarawak-medchain.pages.dev/#/gov-preview
- **Privacy Policy (PDPA 2010):** </privacy>
- **Terms of Service (Malaysian law):** </terms>
- **Public source:** github.com/randyrichard/Sarawak-Medchain

SECTION 3

Market Opportunity

Total Addressable Market (TAM) - Malaysia

~2,000 registered clinics and 380 hospitals (public and private) issue medical certificates. KKM's announced e-MC direction creates a national mandate horizon of 18-36 months.

Serviceable Addressable Market (SAM) - Sarawak (initial focus)

27 hospitals · 142 clinics · ~1,247 registered doctors · population 2.9 million. State-level Sarawak Digital Economy Corporation (SDEC) and TEGAS provide institutional support for locally-built solutions.

Serviceable Obtainable Market (SOM) - 12 months

Q3 2026	1 hospital pilot (target: Sarawak General Hospital, Kuching)
Q4 2026	2-3 additional Sarawak facility pilots
Q1 2027	Convert pilots to paid subscriptions
Q2 2027	5-8 paying facilities across Sarawak

Competitive landscape

KKM's future e-MC system

Likely 18-36 months out. **Sarawak MedChain's advantage is being ready today** - while KKM is still studying, Sarawak can pilot first.

Private EMR vendors (Cerner, Epic)

Priced for national tenders (RM 10M+); not accessible to Sarawak's smaller facilities.

Foreign blockchain healthcare startups

None are Malaysia-focused, PDPA-compliant, or Sarawak-based.

SECTION 4

Business Model

Revenue streams

- **SaaS subscription** - Monthly per-clinic/per-hospital tier
- **Per-MC issuance credits** - Pre-paid credits for high-volume issuers
- **Enterprise / government tier** - Custom pricing for state agencies and large hospital groups

Pricing (base subscription + per-MC usage)

Tier	Target facility	Base / month	Per MC
Clinic	Klinik swasta, GP	RM 2,000	RM 1
Hospital	Private + govt hospitals	RM 10,000+	RM 1
Government	State agencies (SDEC / KKM)	Custom (from RM 10,000)	Custom

Unit economics (12-month projection)

Target by month 12: 5 paying clinics + 2 hospitals.

Revenue line	Monthly
5 clinics x RM 2,000 base subscription	RM 10,000
2 hospitals x RM 10,000 base subscription	RM 20,000
5 clinics x ~500 MCs/mo x RM 1 (usage)	RM 2,500
2 hospitals x ~5,000 MCs/mo x RM 1 (usage)	RM 10,000
Total Monthly Recurring Revenue (MRR)	RM 42,500
Annualised Recurring Revenue (ARR)	RM 510,000

Cost to serve: ~15-20% (backend hosting, IPFS pinning, support). **Gross margin:** ~80%. Model scales cleanly - most costs are already sunk in the platform.

Free pilot as customer-acquisition strategy

Every prospective facility receives a **free 30-day audit pilot**: no cost during pilot, formal audit report delivered at the end, no commitment to continue, founder personally supports onboarding. The audit report is the deliverable that converts pilots to paid subscriptions.

SECTION 5

Traction & Roadmap

What's shipped (as of 30 June 2026)

- Live product across 12+ polished public pages
- Working OTP access demo (patient generates 6-digit code → doctor enters → sees record)
- Full stack: React/Vite frontend on Cloudflare Pages, Node.js/Express backend, Ethereum-compatible smart contract, IPFS storage
- PDPA-compliant Privacy Policy and Terms of Service pages
- Astro Awani citation for the Minister's e-MC announcement embedded in the site
- 1-page pitch PDFs (general and Cradle-tailored versions)
- 30-day pilot audit report template
- Public source code on GitHub

Business milestones (June 2026)

- **SSM registration:** Medchain Enterprise registered June 2026
- **Cradle LIVE! attendance:** attended TEGAS Miri session, 30 June 2026
- **Warm intros in progress:** SDEC, community leadership, YB-level connections via community network

12-month roadmap

Period	Milestone
Q3 2026 Jul - Sep	First paid pilot signed. Deploy production backend (Render/Fly.io). Real cross-device OTP infrastructure live.
Q4 2026 Oct - Dec	2-3 more pilots active. First hire (part-time BD). Sdn Bhd conversion upon first signed contract.
Q1 2027 Jan - Mar	Convert pilots to paid SaaS. Publish first public audit reports (with facility consent).
Q2 2027 Apr - Jun	5+ paying Sarawak facilities. Expand pricing to insurance verification API.

SECTION 6

Team

Randy Richard - Founder & Sole Developer

- 19 years old, from Miri, Sarawak
- Self-taught software engineer
- Built Sarawak MedChain solo over 7 months
- Shipped a working OTP demo within 24 hours of receiving community feedback
- Handles: product design, frontend, backend, smart contract, deployment, business development

Hiring plan (post-first-contract)

- **Q3-Q4 2026:** Part-time Business Development contractor (Sarawak-based) - RM 3,000/month
- **Q1 2027 onwards:** Evaluate second technical hire (backend/DevOps)
- **Governance:** Convert to Sdn Bhd once first paid contract closes; add advisory board

Advisory network (informal, in development)

Community leaders, state agency contacts, and peer founder relationships (via Cradle LIVE! Sarawak session) form the founder's early advisory perimeter. Formal advisory board planned post-Sdn Bhd conversion.

SECTION 7

Financials

Current state

- **Revenue:** Pre-revenue (intentional - free pilots as customer-acquisition)
- **Funding to date:** Bootstrapped from personal funds
- **Operating costs:** ~RM 500/month (domain, hosting, minimal expenses)
- **Founder runway:** 3-4 months at current burn

The Ask - RM 50,000 to RM 150,000

Programme fit: Cradle CIP Spark or MyStartup. Purpose: 12-month runway to first paying customers.

Use of funds (RM 150,000 target allocation)

Bucket	Amount	Purpose
Production backend	RM 40,000	12 months of Render/Fly.io + Redis + Postgres, IPFS pinning, monitoring
Founder runway	RM 30,000	4 months personal runway (RM 7,500/month) - 100% focus on pilots
BD contractor	RM 25,000	Part-time Sarawak-based BD hire, ~6 months
Sdn Bhd + legal	RM 15,000	Registration + Year-1 company secretary + basic legal + accounting
SGH pilot buffer	RM 20,000	Deployment costs + travel + on-site support for first paid pilot
Marketing & travel	RM 20,000	Travel to Sarawak facilities, printed materials, pitch events
Total	RM 150,000	12-month runway to 5 paying customers

Sensitivity: RM 50,000 (minimum viable ask)

If funded at CIP Spark minimum (RM 50k), we prioritise: production backend (RM 25k) + 3 months founder runway (RM 22.5k) + basic legal (RM 2.5k). Sacrifice: BD contractor, marketing budget.

SECTION 8

Risk & Mitigation

Risk	Mitigation
KKM picks a KL vendor before Sarawak pilots	Move fast on SGH POC in Q3 2026. Sarawak-first is our positioning; being live before KKM finishes studying is the moat.
Non-tech users (elderly patients, guest doctors) can't use wallets	Already shipped: OTP access mode. No app install required for doctor side.
Blockchain "right to be forgotten" concerns under PDPA	Documented in Privacy Policy Section 8: encrypted files can be unpinned + keys destroyed, rendering the on-chain hash meaningless.
Single founder / bus factor	Codebase is open-source; documentation being written; BD contractor hire in Q3 to spread ops load.
Security incident	Client-side encryption + no server-side master key means a breach cannot decrypt records. Regular security review post-Series A.
Adoption resistance from hospital IT	Interoperability by design. Public verification API means agencies can plug into our source of truth for their existing MC checks - one tamper-proof endpoint, no duplicate data entry, no fork in the audit trail.

SECTION 9

Why Cradle / Sarawak Now

1. Timing

The Minister's e-MC announcement is 9 days old. This funding window is time-limited.

2. Local advantage

A Sarawak founder building in Sarawak for Sarawak. TEGAS and SDEC are ready to support Sarawak-first digital wins.

3. Political alignment

MP Sibuti (former Deputy Health Minister) is Sarawakian; the e-MC direction aligns with his prior portfolio.

4. Fast-shipping demonstrated

7-month solo build, working product, real citations, no vaporware.

5. Free pilot offer already public

Waiting on the funded runway to execute at scale.

APPENDIX A

Legal & Compliance

- **Legal entity:** Medchain Enterprise (SSM-registered June 2026)
- **Trading name:** Sarawak MedChain
- **Data residency:** All personal data hosted in Malaysia
- **Regulatory basis:** PDPA 2010 (Act 709) compliant Privacy Policy published
- **Governing law:** Terms of Service under Malaysian law, jurisdiction Kuching, Sarawak

APPENDIX B

Working URLs (verify live)

- **Landing:** <https://sarawak-medchain.pages.dev>
- **OTP demo:** <https://sarawak-medchain.pages.dev/#/otp>
- **Government preview:** <https://sarawak-medchain.pages.dev/#/gov-preview>
- **For Hospitals:** <https://sarawak-medchain.pages.dev/#/pitch>

- **Privacy Policy:** <https://sarawak-medchain.pages.dev/#!/privacy>
 - **Terms of Service:** <https://sarawak-medchain.pages.dev/#!/terms>
 - **GitHub:** <https://github.com/randyrichard/Sarawak-Medchain>
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Prepared 30 June 2026 - Version 1.0 - Randy Richard, Founder